Heeseung Kim

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EXPERIENCE

University of Seoul, Seoul, Korea

- Assistant Professor, Department of Artificial Intelligence (Director, Speech & Interactive AI Lab) Mar 2026 – Present

Qualcomm AI Research Korea, Seoul, Korea

- Senior Research Engineer — Sep 2025 – Feb 2026

EDUCATION

Seoul National University, Seoul, Korea

- Ph.D. in Electrical & Computer Engineering — Mar 2019 – Aug 2025
  - Focus: Deep Learning, Generative Models, Speech LLM, Spoken Dialog Model, Speech Synthesis
  - Cumulative GPA: 3.93 / 4.3
- B.S. in Electrical & Computer Engineering — Mar 2015 – Feb 2019
  - Focus: Signal Processing, Machine Learning, Deep Learning
  - Cumulative GPA: 3.85 / 4.3 (Cum Laude)

RESEARCH INTERESTS

Full-Duplex Speech LLMs, with a focus on extending them toward:

- Omni-Modal Agents (Egocentric Understanding for Smart Glasses, Talking Face Generation)
- Multi-Turn Tool Calling for Voice Agents
- Diffusion-Based LLM Architectures

SELECTED PUBLICATIONS

[1] D. Lee*, Y. Cho*, S. Park, H. Kim†, and S. Yoon†, "**NaturalFlow: Reducing Disruptive Pauses for Natural Speech Flow in Simultaneous Speech-to-Speech Translation**," in ***INTERSPEECH 2026, Long Paper, Oral Presentation***, Jun 2026.

[2] D. Lee*, E. Song, C. Lee, H. Kim†, and S. Yoon†, "**Still Between Us? Evaluating and Improving Voice Assistant Robustness to Third-Party Interruptions**," in ***ACL 2026***, Apr 2026.

[3] H. Kim*, C. Lee*, S. Park, J. Yeom, N. Park, S. Yu, and S. Yoon†, "**Does Your Voice Assistant Remember? Analyzing Conversational Context Recall and Utilization in Voice Interaction Models**," ***ACL 2025 Findings***, Jul 2025.

[4] H. Kim*, S. Seo, K. Jeong, O. Kwon, S. Kim, J. Kim, J. Lee, E. Song, M. Oh, J. Ha, S. Yoon†, and K. Yoo†, "**Paralinguistics-Aware Speech-Empowered Large Language Models for Natural Conversation**," in ***NeurIPS 2024***, Vancouver, Canada, Dec 2024.

[5] H. Kim*, S. Kim*, and S. Yoon†, "**Guided-TTS: A Diffusion Model for Text-to-Speech via Classifier Guidance**," in ***ICML 2022***, Baltimore, Maryland USA, Jul 2022.

RESEARCH EXPERIENCE

Qualcomm AI Research Korea — Sep 2025 – Feb 2026

- **Role**: Senior Research Engineer
- **Project Overview**:
  - **2025.09 – 2026.02**: Developed **Full-Duplex Voice Assistant** enabling natural and simultaneous speech interaction. Improved Full-Duplex Speech LLM performance by leveraging synthetic full-duplex text data.

Naver & Seoul National University Collaboration — Mar 2022 – Jan 2025

- **Role**: Industry-academia collaboration, Electrical & Computer Engineering (Ph.D. Candidate)
- **Project Overview**:
  - **2022.03 – 2023.07**: Built a speaker-adaptive TTS model for more natural voice generation. Incorporated speech input into a GPT-2 scale model by attaching an encoder for ASR (Automatic Speech Recognition) and SER (Speech Emotion Recognition), aiming to enrich speech understanding capabilities.
  - **2023.08 – 2024.05**: Developed an *English spoken dialog system* with an end-to-end pipeline and paralinguistic awareness. This research led to a publication at **NeurIPS 2024**, while simultaneously laying the groundwork for Naver's proprietary Speech LLM.
  - **2024.06 – 2025.01**: Focused on developing a text-to-speech model for synthetic spoken dialog generation.

OPEN-SOURCE

UnitSpeech ★130+ : Official implementation of our INTERSPEECH 2023 paper. (Kim et al., 2023)
NAVER USDM ★90+ : Official implementation of our NeurIPS 2024 paper. (Kim et al., 2024)

INVITED TALKS & LECTURES

Lecture Series (3 days): Speech Processing, Recognition, and Synthesis, Samsung AI Expert Program, Samsung Electronics, 2026
Guest Lecture, SW Leadership and Entrepreneurship Seminar, Ewha Womans University, 2026

"Hey Siri: The Present and Future of Voice Assistants", AGILE, Ewha Womans University, 2026
"Speech Synthesis to Voice Assistant", Supertone, 2025
"Latest Trends in Spoken Dialog Models and Voice Agents", Qualcomm, 2024
"Integrating Paralinguistics in Speech-Empowered Large Language Models for Natural Conversation", HMG Tech. Summit, 2024
"Speech and Spoken Dialog Modeling", Neosapience, 2024
"A case study of research and development at Seoul National University using Amazon Mechanical Turk", AWS Summit Seoul, 2024
"Guided-TTS: A Diffusion Model for Text-to-Speech via Classifier Guidance", Kakao Enterprise, 2022

HONORS & GRANTS

NVIDIA Academic Grant Program (Project "OmniDuplex": 4x RTX PRO 6000 donated to UOS), 2026
Gold Reviewer, International Conference on Machine Learning (ICML), 2026
Top Reviewer, Conference on Neural Information Processing Systems (NeurIPS), 2024
Best Paper Award, ACML 2023, 2023
Best Poster Award, 2022 AIIS Fall Retreat, 2022
Outstanding Paper Award, Hyundai AI Consortium, 2022
Cum Laude, Seoul National University, 2019
Academic Performance Scholarship, Seoul National University: 2016-1, 2018-1,2

SERVICES

Reviewer: NeurIPS 2024-2026, ICML 2025-2026, ICLR 2024-2026, CVPR 2025, ACL Rolling Review (ARR) 2025-2026, IEEE Transactions On Multimedia 2025, ACM Multimedia, AAAI, and more

LANGUAGES

- Korean: Native language.
- English: Intermediate (speaking, reading, writing).

OTHER PUBLICATIONS

CONFERENCES (OTHERS)

[1] J. Moon*, S. Kim, Y. Lee, H. Ahn, S. Park, H. Kim, and K. Shim†, "DELTA-TTS: Adapting Autoregressive Model into Diffusion Language Model for Text-to-Speech," ***ICML 2026 Workshop on Structured Probabilistic Inference & Generative Modeling (SPIGM)***, Jul 2026.

[2] C. Lee*, H. Kim†, and S. Yoon†, "From Awareness to Adherence: Bridging the Context Gap in Spoken Dialogue Systems via Context-Aware Decoding," ***INTERSPEECH 2026***, Jun 2026.

[3] J. Choi*, C. Shin*, Y. Oh, H. Kim, J. Lee, S. Yoon†, "Style-Friendly SNR Sampler for Style-Driven Generation," ***WACV 2026***, Mar 2026.

[4] C. Lee*, H. Kim, J. Yeom, and S. Yoon†, "EdiText: Controllable Coarse-to-Fine Text Editing with Diffusion Language Models," ***ACL 2025***, Jul 2025.

[5] N. Park*, H. Kim, C. Lee, J. Choi, J. Yeom, S. Yoon†, "NanoVoice: Efficient Speaker-Adaptive Text-to-Speech for Multiple Speakers," ***ICASSP 2025, Oral Presentation***, Hyderabad, India, Apr 2025.

[6] J. Yeom*, H. Kim, J. Choi, C. Lee, N. Park, S. Yoon†, "VoiceGuider: Enhancing Out-of-Domain Performance in Parameter-Efficient Speaker-Adaptive Text-to-Speech via Autoguidance," ***ICASSP 2025***, Hyderabad, India, Apr 2025.

[7] C. Shin*, J. Choi, H. Kim, S. Yoon†, "Large-Scale Text-to-Image Model with Inpainting is a Zero-Shot Subject-Driven Image Generator," ***CVPR 2025***, Nashville, Tennessee USA, Jun 2025.

[8] H. Kim*, S. Lee, J. Yeom, C. Lee, S. Kim, and S. Yoon†, "VoiceTailor: Lightweight Plug-In Adapter for Diffusion-Based Personalized Text-to-Speech," in ***INTERSPEECH 2024***, Kos Island, Greece, Sep 2024.

[9] C. Shin*, H. Kim*, C. Lee, S. Lee, and S. Yoon†, "Edit-A-Video: Single Video Editing with Object-Aware Consistency," ***ACML 2023, Oral Presentation, Best Paper Award***, Istanbul, Turkey, Nov 2023.

[10] H. Kim*, S. Kim, J. Yeom, and S. Yoon†, "UnitSpeech: Speaker-adaptive Speech Synthesis with Untranscribed Data," in ***INTERSPEECH 2023, Oral Presentation***, Dublin, Ireland, Aug 2023.

[11] S. Lee*, H. Kim, C. Shin, X. Tan†, C. Liu, Q. Meng, T. Qin, W. Chen, S. Yoon†, and T. Liu, "PriorGrad: Improving Conditional Denoising Diffusion Models with Data-Dependent Adaptive Prior," ***ICLR 2022*** (Virtual), Apr 2022.

[12] U. Hwang*, H. Kim, D. Jung, H. Jang, H. Lee, and S. Yoon†, "Stein Latent Optimization for Generative Adversarial Networks," ***ICLR 2022*** (Virtual), Apr 2022.

[13] S. Yu*, J. Song, H. Kim, S. Lee, W. Ryu, and S. Yoon†, "Rare Tokens Degenerate All Tokens: Improving Neural Text Generation via Adaptive Gradient Gating for Rare Token Embeddings," ***ACL 2022***, Dublin, Ireland, May 2022.

JOURNALS

[1] N. Park*, C. Lee, J. Yeom, H. Kim, and S. Yoon†, "Beyond Language-Specific Neurons: The Challenge of Identifying Speech-Specific Neurons in Multimodal LLMs," ***IEEE Journal of Selected Topics in Signal Processing***, in press.

[2] H. Yoo*, E. Kim*, J. Chung*, H. Cho, S. Jeong, H. Kim, D. Jang, H. Kim, J. Yoon, G. Lee, H. Kang, J. Kim, Y. Yun†, S. Yoon†, Y. Hong, "Silent Speech Recognition with Strain Sensors and Deep Learning Analysis of Directional Facial Muscle Movement," ***ACS Appl. Mater. Interfaces 2022***, Nov 2022.
(Impact Factor: 9.229)

ARXIV (OTHERS)

[1] Y. Lee*, J. Moon, H. Kim, H. Choi, H. Kim, and K. Shim†, "TLDR: Compressing Audio Tokens for Efficient Autoregressive Text-to-Speech," *arXiv:2606.09019*, Jun 2026.

[2] Y. Oh*, D. Lee, S. Park, H. Kim, and S. Yoon†, "Omni-Persona: Systematic Benchmarking and Improving Omnimodal Personalization," *arXiv:2605.09996*, May 2026.

[3] J. Moon*, H. Choi, H. Park, H. Kim, and K. Shim†, "Mask2Flow-TSE: Two-Stage Target Speaker Extraction with Masking and Flow Matching," *arXiv:2603.12837*, Mar 2026.

[4] S. Kim*, H. Kim*, and S. Yoon†, "Guided-TTS 2: A Diffusion Model for High-quality Adaptive Text-to-Speech with Untranscribed Data," *arXiv:2205.15370*, May 2022.

[5] HyperCLOVA X Team, "HyperCLOVA X Technical Report," *arXiv:2404.01954*, Apr 2024.

PATENTS

[1] "Speech recognition using facial skin strain data", S. Yoon, E. Kim, H. Kim. US Patent US11810549B2 (2021) & KR Patent KR20220118583A (2021)

[2] "Method and apparatus for training an unsupervised conditional generative model", S. Yoon, U. Hwang, H. Kim. US Patent US20230394319A1 (2023) & KR Patent KR20230168128A (2023)

(: First author; co-first authors share the mark. †: Corresponding author)*